

SAN DIEGO STATE UNIVERSITY

Phase Diagram Lab

ME 241 Materials Laboratory

Group B

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Abstract

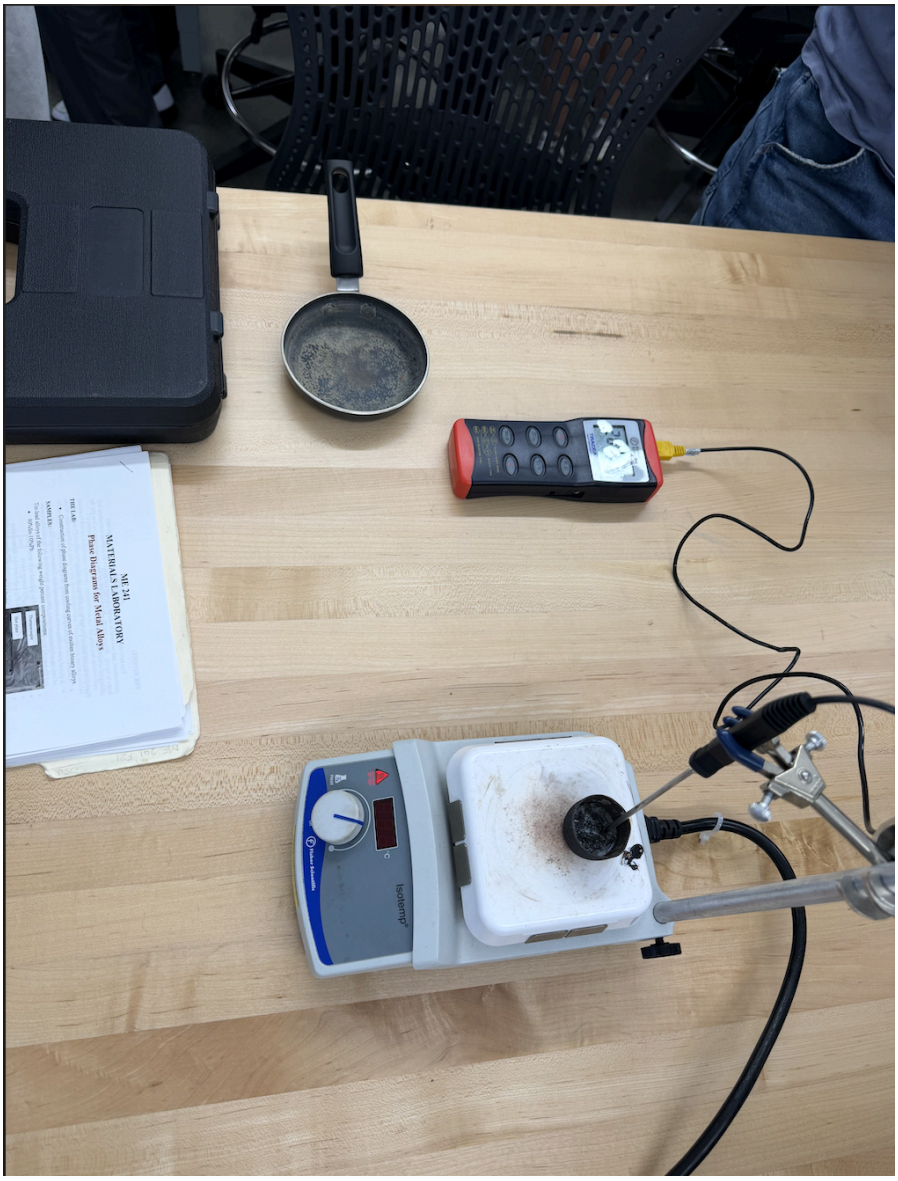
The purpose of this experiment is to understand how a 90% tin–10% lead alloy solidifies by measuring its temperature as it cools. Plotting temperature against time gives us a cooling curve that shows when solidification begins and when the remaining liquid freezes. As the alloy turns solid, it releases latent heat, which temporarily slows the temperature drop. This produces changes in the slope of the curve that we can use to identify phase transitions. The experiment connects these changes to the Sn–Pb phase diagram and helps explain how an alloy's freezing behavior affects processes such as soldering and casting.

Objective

The objective is to create a cooling curve for a 90% Sn–10% Pb alloy, estimate its liquidus and eutectic temperatures, and compare them with the Sn–Pb phase diagram. We will also use the diagram to understand how our sample's composition differs from the eutectic composition. Because this sample is not at the eutectic composition, it is expected to solidify over a range of temperatures. We expect two main features on the cooling curve. The first is a change in slope near the liquidus temperature, where the tin-rich β phase begins to form. The second is a more pronounced leveling of the curve near the eutectic temperature, where the remaining liquid freezes into a mixture of tin-rich and lead-rich solid phases. Comparing the measured temperatures with the phase diagram will also help us assess possible effects of undercooling.

Experimental Equipment

- Safety Glasses
    - For general eye protection; worn by everyone before the hot plate was turned on.
- Stopwatch/timer
    - To keep track of the time elapsed during the experiment.
- Steel crucible
    - A container in which metals or other substances may be melted or subjected to very high temperatures.
- Hot plate
    - A device with a flat heated surface, used to raise the temperature of objects when an open flame is not suitable.
- Thermocouple with calibrated digital display
    - A thermoelectric device for measuring temperature, consisting of two wires of different metals joined at two points, where a voltage develops between the junctions in proportion to the temperature difference.



(Experimental setup)

Procedure & Observations

This experiment involves working close to liquid metal, so proper PPE has to be worn the whole time, including safety glasses and closed-toed shoes.

1) Before starting, assign members of the team to the following:
    a) Monitor the stopwatch and announce the ten-second intervals.
    b) Monitor the thermocouple and announce the temperature reading.

c) Record the temperature reading with respect to the time.

1)Heat the sample above 260 °C until the alloy is uniformly molten, then turn the hot plate off.

2) Record the temperature of the sample at ten-second intervals as it cools. Leave the thermocouple alone once recording starts so its immersion depth does not change partway through.
3) Continue until the thermocouple reads below 88 °C, at which point the alloy is solid.
4) Plot the data as temperature (°C) against time using plotting software.

Our run took 930 seconds and ended at 85.0 °C. We missed the reading at 00:00, so the record starts at 00:10 and 257.1 °C and has 93 points in it. The full table is in the appendix.

Data & Results

The cooling curve was consistent with the expected solidification behavior of a hypereutectic Pb–Sn alloy. Between 00:10 and 02:30, the liquid sample cooled at approximately 0.334 °C/s. The temperature continued to decrease, reaching 207.7 °C at 02:40, before increasing to 210.4 °C at 03:00. This temperature recovery indicates undercooling followed by recalescence, during which solidification releases latent heat. The recovered temperature of 210.4 °C was used as the experimental estimate of the liquidus temperature. During the interval from 02:40 to 03:00, the temperature increased by 2.7 °C, corresponding to an average warming rate of 0.135 °C/s.

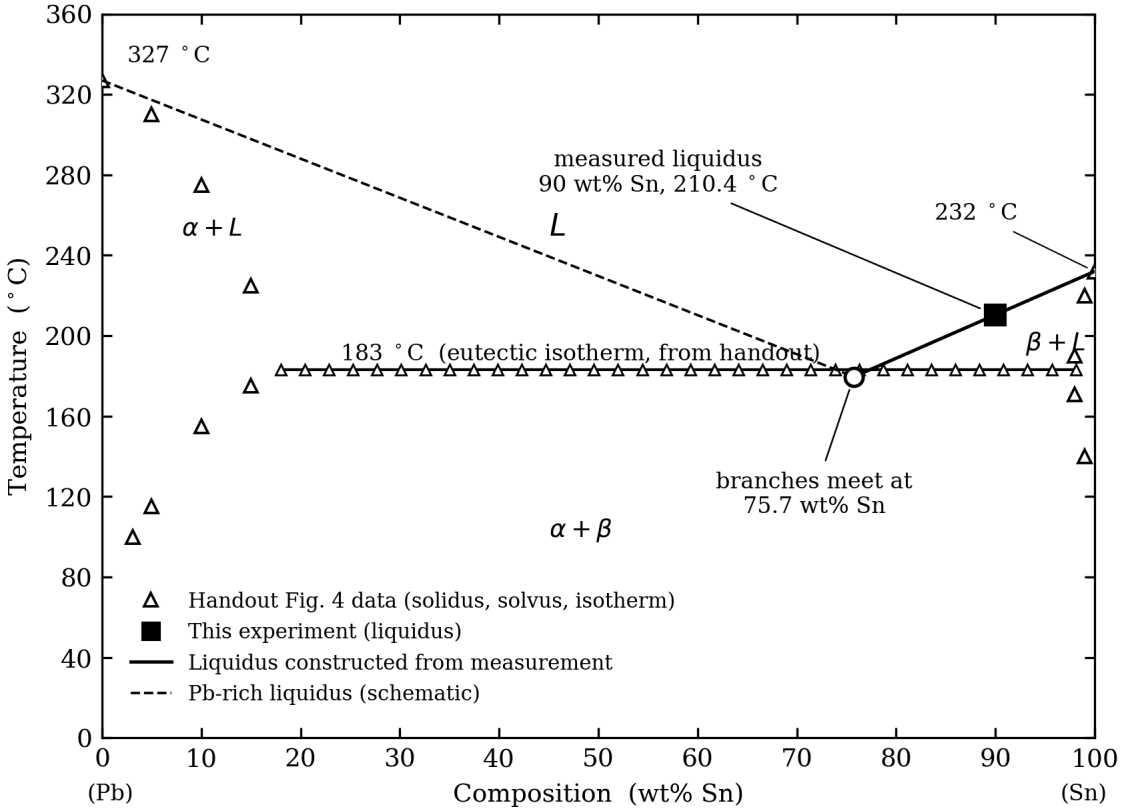
A second region of reduced cooling rate occurred approximately between 06:55 and 07:55. The four temperature readings recorded between 07:10 and 07:40 produced a mean of 179.6 °C, with a sample standard deviation of approximately 0.4 °C and a total range of 0.8 °C.

This region was used to identify the experimental eutectic temperature. Following the completion of solidification at approximately 08:00, the sample resumed cooling more rapidly. The reported cooling rate of 0.317 °C/s was comparable to the initial cooling rate of the liquid sample.

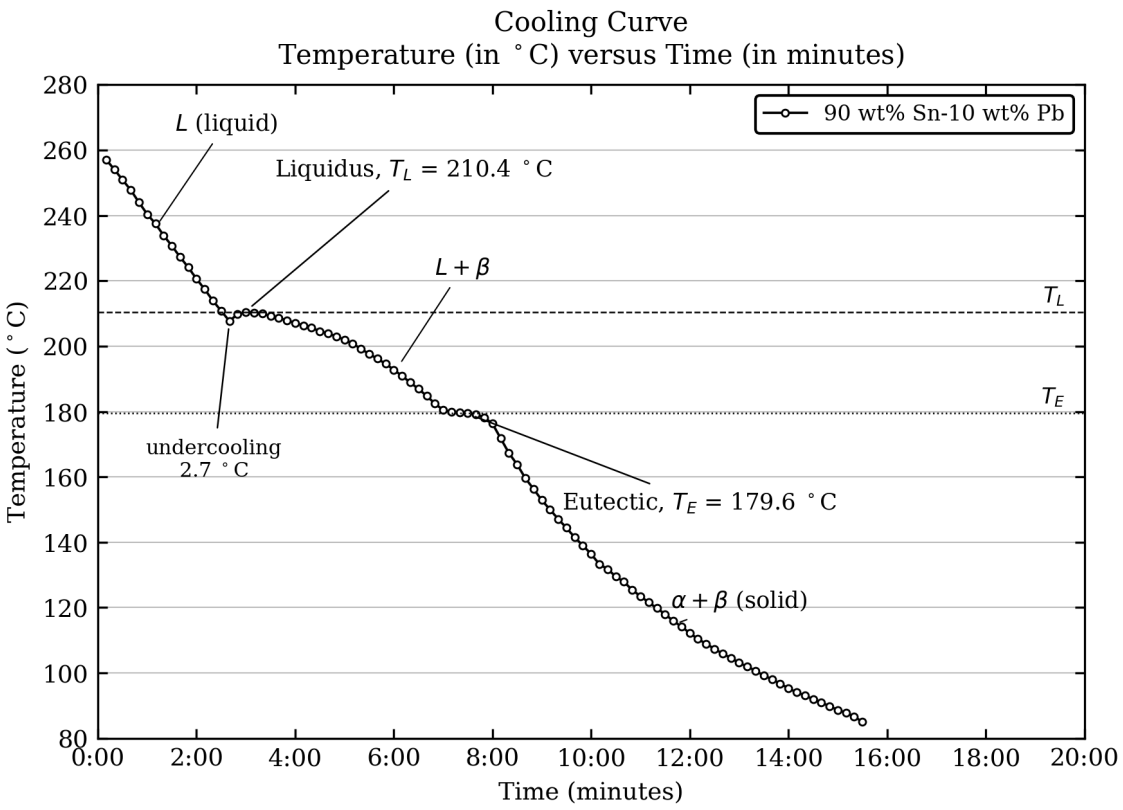
The digital display had a resolution of 0.1 °C, while the stated error limit of the type-K thermocouple was ±2.2 °C. The display resolution therefore should not be interpreted as the accuracy of the temperature measurements. In addition, the temperature changed by approximately 3.3 °C between successive ten-second readings during the initial cooling region. This sampling interval limited the precision with which changes in the cooling curve could be located. Both instrument accuracy and sampling resolution must therefore be considered when interpreting the measured transition temperatures.

Measured values against the published diagram:

Quantity	Measured	Published	Error
Liquidus temperature	210.4 °C	219 °C	−3.93 %
Eutectic temperature	179.6 °C	183 °C	−1.86 %
Eutectic composition	≈ 76 wt% Sn	61.9 wt% Sn	+22 %



(Pb–Sn system phase diagram — handout Fig. 4, completed with our liquidus point)



(Experiment cooling curve for 90 Sn–10 Pb)

Discussion of Results

Both the measured liquidus and eutectic temperatures were lower than the expected values. Potential contributors include thermocouple calibration error, heat conduction through the probe, and temperature variations within the sample. Although the equilibrium eutectic reaction occurs at 183 °C, undercooling can delay the onset of the reaction. Therefore, the 3.4 °C discrepancy cannot be attributed solely to a thermocouple offset without independent calibration.

Assuming a constant thermocouple offset of 3.4 °C below the actual temperature, correcting the measured liquidus would yield 213.8 °C. This remains 5.2 °C below the laboratory reference value of 219 °C. Under this assumption, instrument bias would account for

approximately 40% of the liquidus error. However, this correction remains hypothetical until the thermocouple's calibration is independently verified.

The cooling curve indicated 2.7 °C of undercooling, while the ten-second sampling interval limited the precision with which the onset of solidification could be identified. Near the liquidus break, the estimated temperature change between readings was 3.3 °C. Both effects could contribute to the discrepancy, although their individual contributions cannot be established from the available data. Interpreting the measured liquidus using the phase diagram also suggests a composition closer to 84 wt% Sn. Composition changes associated with repeated melting and dross removal remain a possible explanation, but would require separate verification.

Determination of the eutectic composition was limited by the absence of liquidus data between 15 and 98 wt% Sn in Figure 4. The 90 wt% Sn sample provides only one experimental point within this interval. A straight line connecting the measured liquidus of 210.4 °C at 90 wt% Sn to the melting point of pure tin at 232 °C has a slope of 2.16 °C per wt% Sn. Extrapolating this line to the 183 °C isotherm gives an estimated eutectic composition of 77.3 wt% Sn.

This estimate exceeds the reference composition of 61.9 wt% Sn by approximately 24.9%. Representing the curved liquidus with a straight line contributes to this discrepancy. The low measured liquidus also increases the slope of the fitted line, shifting the calculated composition toward the tin-rich side of the diagram. Consequently, the result should be treated as an approximate extrapolation; the available evidence is insufficient to establish it as an upper bound. Combining results from the other groups would provide additional points across the composition range and improve the determination of the eutectic composition.

Conclusion

The experiment yielded a liquidus temperature of 210.4 °C and a eutectic temperature of 179.6 °C for the 90 Sn–10 Pb alloy. These measurements were 3.93% and 1.86% below the laboratory reference values of 219 °C and 183 °C, respectively. Instrument error, undercooling, and the sampling interval may all have influenced the results, although the experiment does not establish their individual contributions.

For solder applications, the eutectic temperature marks the onset of melting under equilibrium conditions. Using compositions of 97.8 wt% Sn for the β phase and 61.9 wt% Sn for the liquid, the lever rule predicts approximately 22% liquid by mass just above 183 °C, after the eutectic portion has melted. Partial melting can therefore occur while the primary tin-rich crystals remain solid.

The joint does not need to reach its liquidus temperature for melting to affect its performance. Solder can also deform through creep while remaining solid, so 183 °C should not be regarded as a safe operating temperature. An appropriate service temperature must account for the applied load and the duration of thermal exposure.

Future trials should include independent thermocouple calibration, consistent probe immersion depth, and a controlled cooling rate. More frequent temperature measurements near the transitions would improve identification of the cooling curve features. Combining data from multiple groups would also reduce reliance on extrapolation when estimating the eutectic composition.

Appendices

(Data table for the experiment)

Time	Temp. (°C)	Time	Temp. (°C)	Time	Temp. (°C)
00:10	257.1	05:20	199.2	10:30	129.5
00:20	254.1	05:30	197.6	10:40	127.9
00:30	250.9	05:40	196.2	10:50	125.4
00:40	247.8	05:50	194.6	11:00	123.5
00:50	244.0	06:00	192.8	11:10	121.6
01:00	240.4	06:10	190.9	11:20	119.8
01:10	237.5	06:20	188.9	11:30	118.0
01:20	233.9	06:30	186.9	11:40	115.9
01:30	230.7	06:40	184.8	11:50	114.2
01:40	227.3	06:50	182.4	12:00	112.2
01:50	224.1	07:00	180.5	12:10	110.4
02:00	220.6	07:10	179.9	12:20	108.9
02:10	217.4	07:20	179.8	12:30	107.4
02:20	213.9	07:30	179.5	12:40	105.9
02:30	210.8	07:40	179.1	12:50	104.6
02:40	207.7	07:50	178.2	13:00	103.1
02:50	209.8	08:00	176.3	13:10	101.9
03:00	210.4	08:10	171.8	13:20	100.7
03:10	210.3	08:20	167.3	13:30	99.2
03:20	210.0	08:30	163.8	13:40	98.1
03:30	209.2	08:40	159.6	13:50	96.7
03:40	208.6	08:50	156.3	14:00	95.3
03:50	207.9	09:00	152.9	14:10	94.1
04:00	207.1	09:10	150.0	14:20	93.1
04:10	206.3	09:20	147.1	14:30	92.0
04:20	205.6	09:30	144.4	14:40	91.0
04:30	204.6	09:40	141.5	14:50	89.8
04:40	203.9	09:50	139.0	15:00	88.7
04:50	203.0	10:00	136.4	15:10	87.8
05:00	202.0	10:10	133.3	15:20	86.7
05:10	200.8	10:20	131.8	15:30	85.0

No reading was taken at 00:00. Estimated uncertainty: time ±0.5 s, temperature ±2.2 °C.

References

1. Callister, W. D., Jr., and Rethwisch, D. G. Materials Science and Engineering: An Introduction. John Wiley & Sons, Chapter 9, pp. 311–321. ME 241 course excerpt. Lead–tin phase diagram, Figure 9.8; eutectic given as 61.9 wt% Sn at 183 °C. The 219 °C liquidus at 90 wt% Sn was read graphically from that figure.
2. Massalski, T. B., ed. Binary Alloy Phase Diagrams, 2nd ed., Vol. 3. ASM International, Materials Park, OH, 1990. Original source of the diagram reproduced in Reference 1.
3. Higgins, R. Properties of Engineering Materials. Hodder & Stoughton, London, 1986, pp. 149–150. Cooling-curve method as cited in the ME 241 handout.
4. “Phase Diagrams for Metal Alloys.” ME 241 Materials Laboratory handout, Department of Mechanical Engineering, San Diego State University, Fall 2026. Source of the incomplete lead–tin phase diagram (Fig. 4) and the cooling-curve axes (Fig. 3) used in this report.